ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** time period	Time period is measured.
2	**(c)** 0.005 mm	2 / 400 = 0.005 mm.
3	**(d)** rash seen by the doctor	Signs are observed.
4	**(a)** break in the circuit	Loose contact stops current.
5	**(c)** 20 degrees C per minute	Rise of 20 each minute.
6	**(a)** has similar cells doing one job	Organs have more than one tissue type.
7	**(d)** entry, multiplication, symptoms	Pathogen enters and multiplies before symptoms.
8	**(a)** heating	All use resistive heating.
9	**(a)** check the experimental method	Several outliers suggest a faulty procedure.
10	**(d)** provide contrast	Stains help visibility.
11	**(c)** white blood cells and antibodies	Defensive cells and proteins.
12	**(c)** compass needle deflection	Oersted.
13	**(a)** length of the string	Only mass is changed.
14	**(a)** a leaf	Only plant cells have all three.
15	**(b)** contaminated drinking water	Flood mixes sewage with water.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Independent variable is changed by us (1); Dependent variable changes as a result and is measured (1).	1+1
17	They lack a nucleus enclosed by a membrane (1); They are much simpler and smaller (1).	1+1
18	Mosquitoes breed in it (1); Malaria and dengue (1).	1+1
19	Fused filament or loose connection (1); Check connections and try another bulb (1).	1+1
20	Senses can be fooled or are not exact (1); Some things are unsafe to touch or taste (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Change: temperature of water (1); Measure: time taken to dissolve (1); Same: mass of sugar, volume of water, stirring (1).	1+1+1
22	Plant cell (1); Chloroplasts (1); Make food by photosynthesis (1).	1+1+1
23	Vaccine contains weak pathogen or antigen (1); Body makes antibodies and memory cells (1); Boosters keep the memory strong (1).	1+1+1
24	Broken connection or weak cell: replace it (1); No iron core: use an iron nail (1); Too few turns: add turns (1).	1+1+1
	OR: Hand spans differ in size (1); Non-standard units give different values (1); Use a metre scale (standard unit) (1).	1+1+1
25	Actual = 4 / 200 = 0.02 mm (1); 1 mm = 1000 micrometres (1); 0.02 mm = 20 micrometres (1).	1+1+1
	OR: First aid is immediate help before a doctor (1); Medical treatment is by a doctor (1); e.g. cleaning and bandaging a cut (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Measure temperature fall with a thermometer (1); Constant: volume, vessel, starting temperature, place (1); Heat each to same temperature (1); Record every minute in a table (1); Draw a graph and conclude (1).	1+1+1+1+1
	OR: Plant: nucleus, wall, chloroplast, large vacuole (1); Animal: nucleus, no wall, no chloroplast, small vacuoles (1); Bacteria: no true nucleus, wall present, no chloroplast (1); Bacteria are the smallest (1); Bacteria have a nucleoid (1).	1+1+1+1+1

27	Source: infected person's faeces (1); Contaminated water (1); Healthy person drinks it (1); Break: safe water, sanitation (1); Treatment with ORS and care (1).	1+1+1+1+1
	OR: Set-up with wire, cell, switch, compass (1); Needle deflects when current flows (1); Reversing gives opposite deflection (1); Above and below the wire deflections are opposite (1); Conclusion: magnetic field around the wire (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** nichrome	Higher resistance.
2	**(d)** cm	Height is length.
3	**(a)** 10x	400 / 40 = 10.
4	**(b)** before exposure to disease	Preventive.
5	**(b)** the current is switched off	Electromagnet demagnetises.
6	**(c)** height and time	Height changed, time measured.
7	**(a)** 10 mm	0.1 x 100 = 10 mm.
8	**(a)** a physical barrier to germs	Barrier.
9	**(c)** coin, key and iron nail	Metals.
10	**(b)** 12.0 s	(12.1 + 11.9 + 12.0)/3 = 12.0.
11	**(a)** an onion bulb	Underground bulb cells lack chloroplasts but have cell walls.
12	**(c)** stagnant	They breed in still water.
13	**(a)** the switch is open	No current.
14	**(a)** thickness of dough	It is kept same.
15	**(b)** They are large and store water and nutrients	Vacuoles in plants are large.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Prokaryotic: no true nucleus (bacteria) (1); Eukaryotic: true nucleus (plants, animals) (1).	1+1
17	Vaccines prevent disease; antibiotics treat bacterial infection (1); Vaccines train immunity; antibiotics kill bacteria (1).	1+1
18	More turns produce a stronger magnetic field (1); More force on iron pins (1).	1+1
19	A reading far from others (1); Check for error and repeat the measurement (1).	1+1
20	Cytoplasm: jelly-like fluid with organelles (1); Nucleus: has genetic material, controls cell (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	It switches off when current is too large (1); Using a magnet or heating strip (1); Can be reset (1).	1+1+1
22	Why does milk rise on heating? (1); Is it due to bubbles of vapour? (1); Tests can be made with a different liquid (1).	1+1+1
23	A: nerve cell, sends messages (1); B: red blood cell, carries oxygen (1); C: muscle cell, produces movement (1).	1+1+1
24	Protects against tuberculosis (1); Gives immunity early (1); TB is serious (1).	1+1+1
	OR: Cord must carry current with little heat (1); Coil must produce heat (1); Nichrome has high resistance (1).	1+1+1
25	Mean = 54.6 / 3 = 18.2 s (1); Time period = 18.2 / 10 (1); = 1.82 s (1).	1+1+1
	OR: Animal cell (1); Has nucleus and cytoplasm (1); No wall/chloroplast (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Independent: amount of fertiliser (1); Dependent: height of plants (1); Hypothesis: more fertiliser gives more height (1); Keep soil, water, light same (1); Use many plants for each amount (1).	1+1+1+1+1
	OR: Non-cellular, extremely small (1); Multiply inside host cells (1); Diseases: influenza, dengue, measles (1); Prevention: vaccination (1); Hygiene and avoiding crowded places (1).	1+1+1+1+1
27	River water may contain pathogens (1); Causes cholera, typhoid (1); Boil or filter (1); Chlorinate (1); Store in clean covered vessels (1).	1+1+1+1+1

	OR: Fuse and MCB: excess current (1); Earthing: leak current (1); Insulation: shocks (1); Switch on live wire (1); Proper plugs, no overloading (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** monitoring physical health	Physical measurements.
2	**(d)** glows brighter	Greater voltage.
3	**(d)** type of soil	Only the soil type is changed.
4	**(b)** nucleus and cytoplasm	Common parts.
5	**(b)** trapping dust and germs	Sticky layer.
6	**(a)** circles around the wire	Concentric circles.
7	**(d)** 1.8 s	36/20 = 1.8 s.
8	**(c)** They have genetic material but no true nucleus.	Prokaryotic cells.
9	**(d)** immunity	The body remembers the pathogen.
10	**(c)** more clips	More turns, stronger.
11	**(a)** unreliable	Repeatability gives reliability.
12	**(c)** protozoan	Protozoa have a nucleus.
13	**(a)** type 2 diabetes	Linked to diet and exercise.
14	**(d)** heats less due to low resistance	Low resistance.
15	**(c)** How long does ice take to melt in a warm room?	Time must be measured.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Body loses water and salts (1); ORS replaces them and prevents dehydration (1).	1+1
17	Thin wire (1); It has higher resistance (1).	1+1
18	Sum = 16.6 s (1); Mean = 4.15 s (1).	1+1
19	Total = 15 x 20 (1); = 300x (1).	1+1
20	The disease spreads through droplets (1); Isolation breaks the chain of infection (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Provide clean water, chlorinate wells (1); Proper sanitation (1); ORS and health education (1).	1+1+1
22	Thin wires have high resistance and overheat (1); Thick wires carry more current safely (1); Insulation prevents shocks (1).	1+1+1
23	A number without unit has no meaning (1); Prevents mix-up between cm and m (1); Allows other people to repeat the work (1).	1+1+1
24	(a) Chloroplast (1); (b) Nucleus (1); (c) Vacuole (1).	1+1+1
	OR: Illness can cause worry (1); Stress can affect sleep and body (1); Friends' support improves mood (1).	1+1+1
25	Earth wire connects the metal body to the ground (1); Leaks current to earth (1); Prevents shocks (1).	1+1+1
	OR: Same time makes results comparable (1); Fixed procedure removes bias (1); Others can repeat it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Time period rises with length (1); Rise is not proportional (1); Conclusion: longer pendulum, longer period (1); Constant: mass of bob (1); Constant: angle of swing (1).	1+1+1+1+1

	OR: Parts: eyepiece, objective, stage, mirror (1); Light passes through the specimen (1); Lenses magnify the image (1); Total = 12 x 50 (1); = 600x (1).	1+1+1+1+1
27	Skin and mucus: barriers (1); White blood cells engulf germs (1); Antibodies mark germs (1); Memory cells protect later (1); Vaccines use the same principle (1).	1+1+1+1+1
	OR: Coil on soft iron core (1); Current magnetises the core (1); Armature attracted, hammer hits gong (1); Contact breaks, magnet loses power (1); Spring returns, cycle repeats (1).	1+1+1+1+1

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